



PrismLearning.AI

Not just a flashcard generator.

AMD Developer Hackathon

Track 3: Unicorn

Fireworks AI + AMD Instinct™ GPUs

THE PROBLEM

Studying from raw material doesn't scale

- **Sources pile up, structure doesn't**
PDFs, slide decks, and lecture recordings sit in a folder with no connective tissue between them.
- **Flashcard tools stop at the flashcard**
Quizlet and Anki generate cards, then leave you to teach yourself — no explanation, no feedback loop.
- **Generic chatbots answer, they don't teach**
ChatGPT will explain anything once, but it doesn't scaffold understanding, track mastery, or know what you've already learned.

THE SOLUTION

An agentic AI tutor, grounded in your material

Upload a PDF, slide deck, YouTube video, or article and PrismLearning.AI turns it into a structured study workspace with Lumi — an AI tutor that teaches one micro-step at a time, adapts to how you read, and never answers from outside what you gave it.



Agentic AI Tutor

Scaffolded, one micro-step at a time — not a wall of text.



Grounded, Not Generic

Every answer traces back to your own uploaded sources.



Adaptive & Gamified

Reading level, XP, streaks, and boss battles keep it going.



Real Assessments

Math and code questions you actually solve — not just multiple choice.

HOW IT WORKS

From raw source to mastered concept

1 Ingest

Any PDF, slide deck, YouTube video, or article becomes a structured Master Reviewer — a table of contents plus grounded, anchored content.



2 Teach

Lumi walks the Active Learning Canvas with you, one micro-step at a time, checking understanding before advancing.



3 Assess & Reinforce

Quizzes with real math/code questions, spaced-repetition flashcards, and timed boss battles lock in what you learned.

FEATURE HIGHLIGHTS

Built for retention, not just exposure

● Adaptive Reading Levels

Academic → Standard → ELI5, rewritten live by the model, not word-swapped.

● Real Math & Code Quizzes

LaTeX-rendered equations and syntax-highlighted code — problems you solve, not just recognize.

● Spaced Repetition

SM-2-style flashcard scheduling that reviews right before you'd forget.

● Gamification

XP, streaks, mastery decay, and timed Boss Battles for full-chapter mastery.

● Fog-of-War Canvas

Chapters unlock as you actually learn them — no wall-of-text overwhelm.

● Teach-It-Back

The protégé effect: periodically asked to explain concepts in your own words.

MULTI-MODEL ARCHITECTURE

The right model for the right job



gpt-oss-120b

AI Tutoring

Fireworks AI Serverless

Multi-turn scaffolded reasoning — the model that actually teaches Lumi's lessons.



Gemma 3 27B

Flashcard Generation

Fireworks AI Serverless

A short, templated extraction task — deliberately matched to a smaller model, not the biggest one available.



Gemma 4

Enterprise Deployments

AMD Instinct™ GPUs

Self-hosted on AMD Developer Cloud — full data residency for organizations that need it.

Every AI response in this app runs on Fireworks AI's serverless inference platform, powered by AMD Instinct™ GPUs — 100% of generation, zero local/cached/hardcoded output.

Production-shaped, not a hackathon toy



Frontend

- Next.js 14 (App Router) · TypeScript
- Tailwind CSS · shadcn/ui · Framer Motion
- Zustand (agentic UI state) · Mermaid.js
- Clerk (auth)



Backend

- Python 3.11 · FastAPI · Pydantic v2
- openai SDK pointed at Fireworks AI
- python-pptx / pdfplumber / youtube-transcript-api
- Supabase (PostgreSQL + Row-Level Security)



AI & Infra

- gpt-oss-120b · Gemma 3 27B — Fireworks AI Serverless
- Gemma 4 — AMD Instinct™ GPUs (AMD Developer Cloud)
- Supabase · Clerk · Docker Compose
- Vercel (frontend) · Render (backend)

COMPETITIVE LANDSCAPE

Where PrismLearning.AI actually differs

	PrismLearning.AI	Quizlet / Anki	Generic AI Chat	NotebookLM
Grounded in your own material	✓	—	—	✓
Actually teaches (scaffolding)	✓	—	—	—
Adapts reading level live	✓	—	—	—
Real math & code assessment	✓	—	Partial	—
Gamification (XP, streaks, boss fights)	✓	—	—	—
Any-source ingestion (PDF/slides/video/web)	✓	—	—	Partial

BUSINESS MODEL

Simple, fair pricing

Free

Get started

\$0 forever

- ✓ 3 workspaces
- ✓ Core AI tutor
- ✓ Standard quizzes & flashcards

MOST POPULAR

Pro

For serious learners

\$10 / month

- ✓ Unlimited workspaces
- ✓ Math & code question types
- ✓ Boss battles & full gamification
- ✓ Priority AI responses

Enterprise

For schools & teams

Custom

- ✓ Dedicated Gemma 4 on AMD Instinct™
- ✓ SSO & admin dashboard
- ✓ Bulk seats & dedicated support

WHAT'S NEXT

Roadmap



Enterprise & B2B

Dedicated Gemma 4 on AMD Instinct™ GPUs, SSO, admin analytics.



Live Gamified Learning

Real-time multiplayer boss battles and class-wide study leagues.



Voice-Native Tutoring

Talk through a concept with Lumi, hands-free, on any device.



Mobile & Offline

Review decks on the train — sync mastery when you're back online.

BUILT FOR

AMD × Fireworks AI Hackathon — Track 3: Unicorn

Every AI response in this app — ingestion, tutoring, quizzes, and flashcards — is generated by OpenAI's open-weight gpt-oss-120b model, served through Fireworks AI's serverless inference platform, which is powered by AMD Instinct™ GPUs. 100% of generation runs on this path — nothing is generated locally, cached, or hardcoded, and no other inference provider is used anywhere in this codebase.

gpt-oss-120b · Fireworks AI Serverless

Gemma 3 27B · Fireworks AI Serverless

Gemma 4 · AMD Instinct™ GPUs



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Thank you.

Fireworks AI Serverless

Gemma 3 27B

AMD Instinct™ GPUs